Mathematical foundations of density functional theory for one-dimensional systems

Thiago Carvalho Corso
joint work with A. Laestadius
University of Stuttgart, Institute of Applied Analysis and Numerical Simulation (IANS - NMH)

Workshop DFT - Oslo, April 2026

Introduction

Motivation: the ground-state problem in QM

Quantum Mechanics: Let $\Omega \subset \mathbb{R}^d$ be a physical space, and consider the Schrödinger operator

$$H_N(v, w) = -\Delta + \sum_{i<j}^{N} w(x_i, x_j) + \sum_{j=1}^{N} v(x_j),$$

acting on the N-particle (spinless) fermionic space

$$\mathcal{H}_N(\Omega) := \wedge^N \mathrm{L}^2(\Omega; \mathbb{C})$$

Here, v and w and the external and interaction potential (given by physics of the system).

Motivation: the ground-state problem in QM

Quantum Mechanics: Let $\Omega \subset \mathbb{R}^d$ be a physical space, and consider the Schrödinger operator

$$H_N(v, w) = -\Delta + \sum_{i<j}^{N} w(x_i, x_j) + \sum_{j=1}^{N} v(x_j),$$

acting on the N-particle (spinless) fermionic space

$$\mathcal{H}_N(\Omega) := \wedge^N \mathrm{L}^2(\Omega; \mathbb{C})$$

Here, v and w and the external and interaction potential (given by physics of the system).

Ground-state problem: find minimum energy and minimizer of

$$\mathcal{E}_{\mathrm{GS}}(v, w) = \min_{\Psi \in \mathcal{H}_N} \frac{\langle \Psi, H_N(v, w)\Psi \rangle}{\|\Psi\|_{\mathrm{L}^2}^2}.$$

Motivation: the ground-state problem in QM

Quantum Mechanics: Let $\Omega \subset \mathbb{R}^d$ be a physical space, and consider the Schrödinger operator

$$H_N(v,w) = -\Delta + \sum_{i<j}^{N} w(x_i, x_j) + \sum_{j=1}^{N} v(x_j),$$

acting on the N-particle (spinless) fermionic space

$$\mathcal{H}_N(\Omega) := \wedge^N \mathrm{L}^2(\Omega; \mathbb{C})$$

Here, v and w and the external and interaction potential (given by physics of the system).

Ground-state problem: find minimum energy and minimizer of

$$\mathcal{E}_{\mathrm{GS}}(v,w) = \min_{\Psi \in \mathcal{H}_N} \frac{\langle \Psi, H_N(v,w)\Psi \rangle}{\|\Psi\|^2_{\mathrm{L}^2}}.$$

Problem: curse of dimension!

Kohn-Sham Density Functional Theory

DFT idea: shift the focus to ground-state density

$$\rho_\Psi(x) = N \int_{\Omega^{N-1}} |\Psi(x, x_2, ..., x_N)|^2 \mathrm{d}x_2 ... \mathrm{d}x_N.$$

How?

Kohn-Sham Density Functional Theory

DFT idea: shift the focus to ground-state density

$$\rho_\Psi(x) = N \int_{\Omega^{N-1}} |\Psi(x, x_2, ..., x_N)|^2 \mathrm{d}x_2 ... \mathrm{d}x_N.$$

How? In short: the main underlying idea of the Kohn-Sham (KS) method is that, given an interacting (physical) system of interest,

$$H_N(v, w) = -\Delta + \sum_{i \neq j}^{N} w(x_i, x_j) + \sum_{j=1}^{N} v(x_j),$$

one can reproduce its ground-state density and energy via an *effective* non-interacting system,

$$H_N^{\mathrm{KS}}(v_{\mathrm{eff}}) = -\Delta + \sum_{j=1}^{N} v_{\mathrm{eff}}(x_j).$$

Kohn-Sham Density Functional Theory

DFT idea: shift the focus to ground-state density

$$\rho_\Psi(x) = N \int_{\Omega^{N-1}} |\Psi(x, x_2, ..., x_N)|^2 \mathrm{d}x_2 ... \mathrm{d}x_N .$$

How? In short: the main underlying idea of the Kohn-Sham (KS) method is that, given an interacting (physical) system of interest,

$$H_N(v, w) = -\Delta + \sum_{i \neq j}^{N} w(x_i, x_j) + \sum_{j=1}^{N} v(x_j),$$

one can reproduce its ground-state density and energy via an *effective* non-interacting system,

$$H_N^{\mathrm{KS}}(v_{\mathrm{eff}}) = -\Delta + \sum_{j=1}^{N} v_{\mathrm{eff}}(x_j).$$

How to find v_{eff}?

Mathematical foundations of the Kohn-Sham Scheme?

Take a step back: is this even possible? Is KS-DFT an exact theory as often claimed?

Mathematical foundations of the Kohn-Sham Scheme?

Take a step back: is this even possible? Is KS-DFT an exact theory as often claimed?

Warning: this is not possible for the wave-function or other reduced quantities such as single-particle density matrices!

Mathematical foundations of the Kohn-Sham Scheme?

Take a step back: is this even possible? Is KS-DFT an exact theory as often claimed?

Warning: this is not possible for the wave-function or other reduced quantities such as single-particle density matrices!

To seriously think about this, there are several mathematical issues to be addressed:

(1) (Existence or v-representability) Given an interacting Hamiltonian $H_N(v, w)$, is there an effective potential $v_{\mathrm{KS}}(\rho)$ such that $H_N(v_{\mathrm{KS}}, 0)$ reproduces the same ground-state density?

Mathematical foundations of the Kohn-Sham Scheme?

Take a step back: is this even possible? Is KS-DFT an exact theory as often claimed?

Warning: this is not possible for the wave-function or other reduced quantities such as single-particle density matrices!

To seriously think about this, there are several mathematical issues to be addressed:

(1) (Existence or v-representability) Given an interacting Hamiltonian $H_N(v, w)$, is there an effective potential $v_{\mathrm{KS}}(\rho)$ such that $H_N(v_{\mathrm{KS}}, 0)$ reproduces the same ground-state density?

(2) (Uniqueness or HK theorem) Is the previous potential unique?

Mathematical foundations of the Kohn-Sham Scheme?

Take a step back: is this even possible? Is KS-DFT an exact theory as often claimed?

Warning: this is not possible for the wave-function or other reduced quantities such as single-particle density matrices!

To seriously think about this, there are several mathematical issues to be addressed:

(1) (Existence or v-representability) Given an interacting Hamiltonian $H_N(v, w)$, is there an effective potential $v_{\mathrm{KS}}(\rho)$ such that $H_N(v_{\mathrm{KS}}, 0)$ reproduces the same ground-state density?

(2) (Uniqueness or HK theorem) Is the previous potential unique?

(3) (Differentiability) Is the xc-functional differentiable at all? In which sense does the KS equation holds?

The 1D case:

For the rest of this talk, we focus on *one-dimensional* systems, i.e., $\Omega = I = (0,1) \subset \mathbb{R}$ or $\Omega = \mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Goal of this talk: present answers to the previous questions in this 1D setting.

The 1D case:

For the rest of this talk, we focus on *one-dimensional* systems, i.e., $\Omega = I = (0,1) \subset \mathbb{R}$ or $\Omega = \mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Goal of this talk: present answers to the previous questions in this 1D setting.

Wait, why should we care about the 1D case?

The 1D case:

For the rest of this talk, we focus on *one-dimensional* systems, i.e., $\Omega = I = (0,1) \subset \mathbb{R}$ or $\Omega = \mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Goal of this talk: present answers to the previous questions in this 1D setting.

Wait, why should we care about the 1D case?

The honest answer: because we don't know how to deal with higher dimensions. =[

The 1D case:

For the rest of this talk, we focus on *one-dimensional* systems, i.e., $\Omega = I = (0,1) \subset \mathbb{R}$ or $\Omega = \mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Goal of this talk: present answers to the previous questions in this 1D setting.

Wait, why should we care about the 1D case?

The honest answer: because we don't know how to deal with higher dimensions. =[

But: 1D is not trivial!! For large N and $w \neq 0$, it is a PDE (not an ODE) in a high dimensional space.

Rigorous formulation of KS-DFT in 1D

The setting - distributional potentials

We set $I = (0,1)$ and define the spaces of distributional external potentials:

$$\boxed{\mathcal{V}(I) := \mathrm{H}^{-1}(I;\mathbb{R})}$$

The setting - distributional potentials

We set $I = (0,1)$ and define the spaces of distributional external potentials:

$$\boxed{\mathcal{V}(I) := \mathrm{H}^{-1}(I;\mathbb{R})}$$

What the f...? The point is that the potential can be seen as linear functionals acting on the density

$$\rho_\Psi \mapsto \langle \rho_\Psi, v\rangle \quad \text{(dual pairing)}.$$

Similarly, one can consider distributional interaction potentials w as linear functionals acting on the pair densities $\rho_\Psi^{(2)} \mapsto \langle \rho_\Psi^{(2)}, w\rangle$.

The setting - distributional potentials

We set $I = (0,1)$ and define the spaces of distributional external potentials:

$$\boxed{\mathcal{V}(I) := \mathrm{H}^{-1}(I;\mathbb{R})}$$

What the f...? The point is that the potential can be seen as linear functionals acting on the density

$$\rho_\Psi \mapsto \langle \rho_\Psi, v\rangle \quad \text{(dual pairing).}$$

Similarly, one can consider distributional interaction potentials w as linear functionals acting on the pair densities $\rho_\Psi^{(2)} \mapsto \langle \rho_\Psi^{(2)}, w\rangle$. This allows us to define the N-particle Hamiltonian $H_N(v,w) = -\Delta + \sum_{i\neq j}^N w(x_i,x_j) + \sum_{j=1}^N v(x_i)$ via the quadratic form

$$\langle \Psi, H_N(v,w)\Psi\rangle = \int_{I^N} |\nabla\Psi|^2 \mathrm{d}x + \langle \rho_\Psi, v\rangle + \langle \rho_\Psi^{(2)}, w\rangle, \quad \text{with (form) domain } \mathcal{H}_N \cap \mathrm{H}^1(I^N).$$

The setting - distributional potentials

We set $I = (0,1)$ and define the spaces of distributional external potentials:

$$\boxed{\mathcal{V}(I) := \mathrm{H}^{-1}(I;\mathbb{R})}$$

What the f...? The point is that the potential can be seen as linear functionals acting on the density

$$\rho_\Psi \mapsto \langle \rho_\Psi, v\rangle \quad \text{(dual pairing)}.$$

Similarly, one can consider distributional interaction potentials w as linear functionals acting on the pair densities $\rho_\Psi^{(2)} \mapsto \langle \rho_\Psi^{(2)}, w\rangle$. This allows us to define the N-particle Hamiltonian $H_N(v,w) = -\Delta + \sum_{i\neq j}^N w(x_i,x_j) + \sum_{j=1}^N v(x_i)$ via the quadratic form

$$\langle \Psi, H_N(v,w)\Psi\rangle = \int_{I^N} |\nabla\Psi|^2 \mathrm{d}x + \langle \rho_\Psi, v\rangle + \langle \rho_\Psi^{(2)}, w\rangle, \quad \text{with (form) domain } \mathcal{H}_N \cap \mathrm{H}^1(I^N).$$

Remarks: (1) As $\mathrm{H}^1(I^N)$ is compactly embedded in $\mathrm{L}^2(I^N)$, the spectrum of $H_N(v,w)$ is discrete.
(2) The form domain considered here is also called the Neumann form domain.
(3) This class of potentials appeared for the first time (in this context) in recent work by Sutter et al (JPA 2024), where they provided sufficient conditions for ensemble v-rep on $\mathbb{T}$.

The pure-state v-representability problem

Question (1) - $\mathcal{V}$-representability: can one characterize the space of ground-state densities? In our setting, the following space

$$\mathcal{D}_N(w;\mathcal{V}) \coloneqq \{\rho = \rho_\Psi : \Psi \text{ ground-state of } H_N(v,w) \text{ for some } v \in \mathcal{V}\}.$$

The pure-state v-representability problem

Question (1) - $\mathcal{V}$-representability: can one characterize the space of ground-state densities? In our setting, the following space

$$\mathcal{D}_N(w;\mathcal{V}) := \{\rho = \rho_\Psi : \Psi \text{ ground-state of } H_N(v,w) \text{ for some } v \in \mathcal{V}\}.$$

In our setting, one can obtain a complete solution of this problem.

Theorem (Characterization of pure-state $\mathcal{V}$-representability)

Let $w \in \mathcal{W}(I)$ be fixed and $N \in \mathbb{N}$. Then a function ρ is the density of a ground-state of $H_N(v,w)$ for some $v \in \mathcal{V}(I)$ if and only if ρ belongs to the set

$$\mathcal{D}_N(w;\mathcal{V}) = \left\{\rho : \sqrt{\rho} \in \mathrm{H}^1(I), \quad \int_I \rho(x)\mathrm{d}x = N \quad \textit{and} \quad \rho(x) > 0 \quad \textit{for any } x \in [0,1]\right\}. \qquad (1)$$

The pure-state v-representability problem

Question (1) - $\mathcal{V}$-representability: can one characterize the space of ground-state densities? In our setting, the following space

$$\mathcal{D}_N(w;\mathcal{V}) := \{\rho = \rho_\Psi : \Psi \text{ ground-state of } H_N(v,w) \text{ for some } v \in \mathcal{V}\}.$$

In our setting, one can obtain a complete solution of this problem.

Theorem (Characterization of pure-state $\mathcal{V}$-representability)

Let $w \in \mathcal{W}(I)$ be fixed and $N \in \mathbb{N}$. Then a function ρ is the density of a ground-state of $H_N(v,w)$ for some $v \in \mathcal{V}(I)$ if and only if ρ belongs to the set

$$\mathcal{D}_N(w;\mathcal{V}) = \left\{\rho : \sqrt{\rho} \in \mathrm{H}^1(I), \quad \int_I \rho(x)\mathrm{d}x = N \quad \textit{and} \quad \rho(x) > 0 \quad \textit{for any } x \in [0,1]\right\}. \qquad (1)$$

In particular, $\mathcal{D}_N(w) = \mathcal{D}_N$ is independent of the interaction potential $w \in \mathcal{W}(I)$, and therefore, for every interacting system $H_N(v,w)$, there exists a potential $v_{\mathrm{KS}}(v,w)$ such that $H_N(v_{\mathrm{KS}},0)$ has the *exact* same ground-state density.

What about uniqueness?

Question (2): is the system with ground-state density unique? Do we lose uniqueness by extending the space of potentials to distributions?

What about uniqueness?

Question (2): is the system with ground-state density unique? Do we lose uniqueness by extending the space of potentials to distributions?

Answer: no! In fact, we have something stronger.

Theorem (Quantitative Hohenberg-Kohn theorem)

Let $\rho \in \mathcal{D}_N$, then for any sufficiently small neighborhood $\rho \in U \subset \mathcal{D}_N$, there exists a constant $C > 0$ such that

$$\|v(\rho; w) - v(\rho'; w)\|_{\mathcal{V}/\{1\}} \leq C\|\rho - \rho'\|_{\mathrm{H}^1},$$

for any $\rho' \in U$.

What about uniqueness?

Question (2): is the system with ground-state density unique? Do we lose uniqueness by extending the space of potentials to distributions?

Answer: no! In fact, we have something stronger.

Theorem (Quantitative Hohenberg-Kohn theorem)

Let $\rho \in \mathcal{D}_N$, then for any sufficiently small neighborhood $\rho \in U \subset \mathcal{D}_N$, there exists a constant $C > 0$ such that

$$\|v(\rho; w) - v(\rho'; w)\|_{\mathcal{V}/\{1\}} \leq C\|\rho - \rho'\|_{\mathrm{H}^1},$$

for any $\rho' \in U$.

Remarks: This shows that the inverse problem is Lipschitz stable.

What about uniqueness?

Question (2): is the system with ground-state density unique? Do we lose uniqueness by extending the space of potentials to distributions?

Answer: no! In fact, we have something stronger.

Theorem (Quantitative Hohenberg-Kohn theorem)

Let $\rho \in \mathcal{D}_N$, then for any sufficiently small neighborhood $\rho \in U \subset \mathcal{D}_N$, there exists a constant $C > 0$ such that

$$\|v(\rho; w) - v(\rho'; w)\|_{\mathcal{V}/\{1\}} \leq C\|\rho - \rho'\|_{\mathrm{H}^1},$$

for any $\rho' \in U$.

Remarks: This shows that the inverse problem is Lipschitz stable.
Combining these results we conclude that, for any $w \in \mathcal{W}(I)$, there exists a bijective map

$$\rho \in \mathcal{D}_N \mapsto v(\rho; w) \in \mathcal{V}(I)/\{1\},$$

where $\mathcal{V}(I)/\{1\}$ is the space of equivalence classes in $\mathcal{V}(I)$ modulo additive constants.

How regular is DFT?

Question (3): What about the regularity? Is the exchange-correlation differentiable?

How regular is DFT?

Question (3): What about the regularity? Is the exchange-correlation differentiable?
Yes. In fact, surprisingly (to me), it is not only differentiable!

Theorem (Analytic functional theory)

The constrained search functional

$$F_{\mathrm{LL}}(\rho; w) = \inf_{\Psi \mapsto \rho} \langle \Psi, \hat{T} + \hat{W}, \Psi \rangle$$

is real-analytic on the space $\mathcal{D}_N \times \mathbb{R}$*. In particular, the xc-energy is also analytic and*

$$\mathrm{d}_\rho E_{\mathrm{xc}} = v_{\mathrm{xc}}(\rho) \in \mathcal{V}/\{1\}.$$

How regular is DFT?

Question (3): What about the regularity? Is the exchange-correlation differentiable?
Yes. In fact, surprisingly (to me), it is not only differentiable!

Theorem (Analytic functional theory)

The constrained search functional

$$F_{\mathrm{LL}}(\rho; w) = \inf_{\Psi \mapsto \rho} \langle \Psi, \hat{T} + \hat{W}, \Psi \rangle$$

is real-analytic on the space $\mathcal{D}_N \times \mathbb{R}$. In particular, the xc-energy is also analytic and

$$\mathrm{d}_\rho E_{\mathrm{xc}} = v_{\mathrm{xc}}(\rho) \in \mathcal{V}/\{1\}.$$

Remark: Recalling that, for $\rho \in \mathcal{D}_N$, $T_\rho \mathcal{D}_N = \mathcal{X}_0 = \left\{\rho \in \mathrm{H}^1(I;\mathbb{R}) : \int_I \rho(x)\mathrm{d}x = 0\right\}$, we say that $F : \mathcal{D}_N \times \mathbb{R} \to \mathbb{R}$ is analytic around a point (ρ, λ), if we have a (norm convergent) power series

$$F_{\mathrm{LL}}(\rho + \sigma, \lambda + \mu) = \sum_{n \geq 0} F^{(n)}_{(\rho,\lambda)} \left((\sigma, \mu), \ldots (\sigma, \mu)\right).$$

Applications: prove cool stuff

The KS-scheme is rigorously exact!

We can justify the KS scheme.

Corollary (Exact Kohn-Sham DFT)

Let $v \in \mathcal{V}$, $w \in \mathcal{W}$ and $N \in \mathbb{N}$, and denote by $\rho(v; w) \in \mathcal{D}_N$ the (unique) ground-state density of $H_N(v, w)$. Then, there exists a unique (up to a global phase) minimizer of

$$\min_{\substack{\Psi \in \textit{ Slater} \\ \|\Psi\|=1}} \left\{ \mathcal{E}_{\mathrm{KS}}(\Psi) = T_{\mathrm{KS}}(\rho_\Psi) + E_{\mathrm{xc}}(\rho_\Psi) + E_H(\rho_\Psi) + \langle \rho_\Psi, v \rangle \right\}.$$

Moreover, this minimizer is given by the Slater determinant of the N lowest eigenfunctions of the Kohn-Sham single-particle Hamiltonian

$$h(\rho(v; w)) = -\Delta + v_{\mathrm{xc}}(\rho(v; w)) + v_H(\rho(v; w)) + v,$$

where $v_{\mathrm{xc}}(\rho) = \mathrm{d}_\rho E_{\mathrm{xc}} \in \mathcal{V}$ and $v_H(\rho) \in \mathcal{V}$ is the Hartree (distributional) potential

$$\delta \mapsto v_H(\rho)(\delta) = \langle \rho \otimes \delta, w \rangle + \langle \delta \otimes \rho, w \rangle. \tag{2}$$

Adiabatic connection exists

You can also prove that the adiabatic connection exists!

Corollary (Adiabatic connection)

For any $\rho \in \mathcal{D}_N$ and $\lambda \in \mathbb{R}$ there exists one (and only one) potential $v(\rho, \lambda)$ such that ρ is the ground-state of

$$H_N(v(\rho, \lambda), \lambda w).$$

Moreover, the map

$$(\rho, \lambda) \to v(\rho, \lambda)$$

is analytic in $\mathcal{D}_N \times \mathbb{R}$.

Adiabatic connection exists

You can also prove that the adiabatic connection exists!

Corollary (Adiabatic connection)

For any $\rho \in \mathcal{D}_N$ and $\lambda \in \mathbb{R}$ there exists one (and only one) potential $v(\rho, \lambda)$ such that ρ is the ground-state of

$$H_N(v(\rho, \lambda), \lambda w).$$

Moreover, the map

$$(\rho, \lambda) \to v(\rho, \lambda)$$

is analytic in $\mathcal{D}_N \times \mathbb{R}$.

Remark: Hence, the λ xc-energy

$$E_{\mathrm{xc}}(\rho; \lambda) = F_{\mathrm{LL}}(\rho, \lambda) - F_{\mathrm{LL}}(\rho, 0) - \lambda E_H(\rho)$$

is also analytic in λ.

Exchange-only potential and Görling-Levy perturbation series

Corollary (Existence of exchange-only potential)

The xc-potential $v_{\mathrm{xc}}(\rho;\lambda)$ *is also analytic. In particular,*

$$\left.\frac{\mathrm{d}v_{\mathrm{xc}}(\rho;\lambda)}{\mathrm{d}\lambda}\right|_{\lambda=0} = v_{\mathrm{x}}(\rho;w) \in \mathcal{V}/\{1\} \quad \textit{is well-defined and satisfy} \quad v_{\mathrm{x}}(\rho;w) = D_\rho E_{\mathrm{x}}(\rho;w),$$

where $E_x(\rho;w) = \frac{dE_{\mathrm{xc}}(\rho;\lambda w)}{\mathrm{d}\lambda}|_{\lambda=0}$ *is the exchange energy.*

Exchange-only potential and Görling-Levy perturbation series

Corollary (Existence of exchange-only potential)

The xc-potential $v_{\mathrm{xc}}(\rho;\lambda)$ *is also analytic. In particular,*

$$\left.\frac{\mathrm{d}v_{\mathrm{xc}}(\rho;\lambda)}{\mathrm{d}\lambda}\right|_{\lambda=0} = v_{\mathrm{x}}(\rho;w) \in \mathcal{V}/\{1\} \quad \textit{is well-defined and satisfy} \quad v_{\mathrm{x}}(\rho;w) = D_\rho E_{\mathrm{x}}(\rho;w),$$

where $E_x(\rho;w) = \frac{dE_{\mathrm{xc}}(\rho;\lambda w)}{\mathrm{d}\lambda}|_{\lambda=0}$ *is the exchange energy.*

Corollary (Görling-Levy perturbation series)

The correlation energy $E_c(\rho;\lambda) = E_{\mathrm{xc}}(\rho;\lambda w) - E_x(\rho;\lambda)$ *is also analytic and admits an expansion*

$$E_c(\rho;\lambda) = \sum_{n\geq 2}^{\infty} \lambda^n E_c^{GLn}(\rho;w), \quad \textit{where} \quad E_c^{\mathrm{GL2}}(\rho) = -\sum_{j\geq 1} \frac{|\langle \Psi_0, (\hat{w} - \hat{v}_x(\rho) - \hat{v}_H(\rho))\, \Psi_j\rangle|^2}{\omega_j},$$

and $E_c^{\mathrm{GLn}}(\rho;w)$ *are the higher order terms of the Görling-Levy perturbation series.*

Kantorovich potentials in the strongly interacting limit

One can also justify the (leading order) expansion of the AC potential $v(\rho, \lambda)$ as $\lambda \to \infty$.

Theorem (Leading order expansion of adiabatic potential)

Under suitable continuity and repulsive assumptions on w, for any density $\rho \in \mathcal{D}_N$, we have

$$\frac{v(\rho; \lambda)}{\lambda} \rightharpoonup v_{\mathrm{OT}}(\rho) \quad \textit{in } \mathcal{V}/\{1\},$$

where $v_{\mathrm{OT}}(\rho)$ is a continuous Kantorovich potential of the optimal transport problem

$$F_{\mathrm{OT}}(\rho) = \min \left\{ \int_{I_n} \sum_{j \neq k} w(x_j, x_k) \gamma(\mathrm{d}X) : \gamma \in \mathcal{P}(I_n) \quad (\pi_j^{\#} \gamma) = \frac{\rho}{n}, \quad j = 1, ..., n \right\}.$$

Kantorovich potentials in the strongly interacting limit

One can also justify the (leading order) expansion of the AC potential $v(\rho, \lambda)$ as $\lambda \to \infty$.

Theorem (Leading order expansion of adiabatic potential)

Under suitable continuity and repulsive assumptions on w, for any density $\rho \in \mathcal{D}_N$, we have

$$\frac{v(\rho;\lambda)}{\lambda} \rightharpoonup v_{\mathrm{OT}}(\rho) \quad \text{in } \mathcal{V}/\{1\},$$

where $v_{\mathrm{OT}}(\rho)$ is a continuous Kantorovich potential of the optimal transport problem

$$F_{\mathrm{OT}}(\rho) = \min\left\{ \int_{I_n} \sum_{j\neq k} w(x_j, x_k)\gamma(\mathrm{d}X) : \gamma \in \mathcal{P}(I_n) \quad (\pi_j^{\#}\gamma) = \frac{\rho}{n}, \quad j = 1, ..., n \right\}.$$

In other words:

$$v(\rho;\lambda) = v_{\mathrm{OT}}(\rho)\lambda + o(\lambda) \quad \text{as } \lambda \to \infty.$$

If you like TDDFT?

One can prove the existence and uniqueness of an exact Hartree-exchange-correlation kernel in linear response TDDFT.

Theorem (Exact Hxc-kernel)

There exists a unique (analytic) on the upper half plane functional $F_{w,\rho} : \mathbb{C}_+ \to \mathcal{B}(\mathcal{X}_0, \mathcal{V}/\{1\})$ *such that*

$$\hat{\chi}(z) = \hat{\chi}_0(z) + \hat{\chi}_0(z)\hat{F}_{\mathrm{Hxc}}(z)\hat{\chi}(z),$$

where $\hat{\chi}$ *and* $\hat{\chi}_0$ *are, respectively, the density-response functions of the interacting and non-interacting systems (in the frequency domain).*

A holomorphic extension of DFT

Theorem (Holomorphic extension of DFT)

Exists a rel. open set $U \subset \mathcal{X}_N^{\mathbb{C}} \times \mathbb{C}$ and a (ess. unique) holomorphic function $\widetilde{F}_{\mathrm{LL}} : U \to \mathbb{C}$ s.t.

(i) (Extension) We have $\mathcal{D}_N \times \mathbb{R} \subset U$ and $\widetilde{F}_{\mathrm{LL}}(\rho, \lambda) = F_{\mathrm{LL}}(\rho, \lambda w)$ for any $(\rho, \lambda) \in \mathcal{D}_N \times \mathbb{R}$. Moreover, the extension is essentially unique.

(ii) (Bijective density-to-potential map) The set

$$V := \{(-D_\rho F_{\mathrm{LL}}(\rho, \lambda), \lambda) : (\rho, \lambda) \in U\} \subset \mathrm{H}^{-1}(I; \mathbb{C})/\{1\} \times \mathbb{C},$$

is open and the map $G : U \to V$ given as follows is a bi-holomorphic bijection

$$(\rho, \lambda) \in U \mapsto G(\rho, \lambda) = (v(\rho, \lambda), \lambda) := (-D_\rho F_{\mathrm{LL}}(\rho; \lambda w), \lambda) \in V.$$

(iii) (Potential-to-density map) Exists a projection-valued holomorphic map $P : V \mapsto \mathcal{S}_1^1$ such that $G^{-1}(v, \lambda) = (\rho_{P(v,\lambda)}, \lambda)$, for $(v, \lambda) \in V$, where $\rho_{P(v,\lambda)}$ is the density of $P(v, \lambda)$. Moreover,

$$\left[H_N(v, \lambda w) - \left(\tilde{F}(G^{-1}(v, \lambda)) + \langle \rho(v, \lambda), v \rangle\right) 1\right] P(v, \lambda) = 0, \quad \textit{for } (v, \lambda) \in V.$$

Conclusion

Summary

I tried to convince you that all mathematical subtleties of exact KS-DFT can be addressed in 1D, i.e,

(i) One has a complete solution of the *pure-state* v-representability problem in one-dimensional interval with (generalized) Neumann BCs..

(ii) The (quantitative) HK theorem still holds for distributional potentials.

(iii) The xc-functional is differentiable.

(iv) One can prove lots of stuff.

Summary

I tried to convince you that all mathematical subtleties of exact KS-DFT can be addressed in 1D, i.e,

(i) One has a complete solution of the *pure-state* v-representability problem in one-dimensional interval with (generalized) Neumann BCs..

(ii) The (quantitative) HK theorem still holds for distributional potentials.

(iii) The xc-functional is differentiable.

(iv) One can prove lots of stuff.

All of these were only possible by using the "correct" space of one-body potentials $\mathcal{V}(I) = \mathrm{H}^{-1}(I;\mathbb{R})$.

In particular, this suggests that distributional potentials are not only sufficient but also **necessary** to a consistent (rigorous) formulation of (KS)-DFT.

Many further questions: Dirichlet BCs, spin, the whole space, higher dimensions...

Thanks for the attention!

References:

(∗) T. C. Corso, *v-representability and Hohenberg-Kohn theorem for non-interacting Schrödinger operators with distributional potentials on the one-dimensional torus*, *Journal of Physics A: Mathematical and Theoretical* (2025). https://doi.org/10.1088/1751-8121/adc04c

(∗) T. C. Corso, *A non-degeneracy theorem for interacting fermions in one dimension*, *Annales Henri Poincaré* (2026). https://doi.org/10.1007/s00023-026-01693-0

(∗) T. C. Corso, *A rigorous formulation of density functional theory for spinless fermions in one dimension*, *Letters in Mathematical Physics* (2026). https://doi.org/10.1007/s11005-026-02051-1.

(∗) T. C. Corso and A. Laestadius, *A quantitative Hohenberg-Kohn theorem and the unexpected regularity of DFT in one spatial dimension*, arXiv:2512.04726.

(∗) T. C. Corso, *Strictly correlated electrons in a quantum ring: from Kohn-Sham to Kantorovich potentials*, arXiv:2604.09908

Questions?